

Arjun Nanduri

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Education

University of California: Berkeley, BA in Computer Science

Expected May 2028

- GPA: 3.9 | SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Computer Security, Data Structures

Research

Undergraduate Researcher, Abbasi Lab, UCSF

Sep 2026 – Present

- Applying **unsupervised and semi-supervised deep learning** (deep autoencoders, non-negative matrix factorization (NMF), semi-supervised CNNs) to structural MRI/fMRI from thousands of UCSF patients and public datasets, plus Allen Institute gene expression images, to relate patterns in unlabeled medical images to patient health outcomes.
- Applying **large language model (LLM) interpretability methods** to evaluate the validity of outputs from spatial queries in spatial transcriptomics, testing on smaller datasets and on larger spatial foundation models.

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA

Jan 2025 – May 2025

- Fine-tuned BART transformer for Ancient Akkadian lemmatization in low-resource settings, **beating open-source baselines by 15%** using Levenshtein distance over lemma forms.
- Built preprocessing and alignment pipelines for structured linguistic data from ORACC's cuneiform corpus (**~1M annotated words across ~10K texts**).
- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.

Experience

Machine Learning Engineer, Aver Technologies Inc. – Virginia

Aug 2025 – Present

- Building a multi-stage PyTorch CV pipeline to automate depth tracking when driving piles (foundational support for heavy structures, like bridges) into the ground — **replacing a manual, hand-marked DOT compliance process**.
- Trained a CV pipeline for **real-time pile depth tracking** — YOLO detection and segmentation to track painted depth markings, audio-visual sync to isolate hammer strikes, and depth-map pixel-to-real-world conversion.
- Conducted field interviews with construction workers to surface real-world constraints — inconsistent hand-drawn markings, Spanish-speaking workforce — **directly informing model design decisions**.
- **Chose CV over LIDAR for cost-effectiveness** — helped file a provisional patent, designing a tripod-mounted 1000fps camera and Jetson Orin GPU capture rig.

Software Engineering Intern, CACI International Inc. – Virginia

May 2025 – Aug 2025

- Built a full-stack logistics web app for **U.S. Customs field officers** using Angular, Spring Boot, and PostgreSQL.

Projects

PotBot — Civic Reporting Agent, CalHacks Prize Winner

<https://devpost.com/software/potbot>

- **Won sponsor prize at CalHacks** — built agentic pipeline using Llama vision model to validate infrastructure damage photos and auto-submit repair requests to SF city government via Playwright browser automation.

Quadruped Locomotion with PPO (MuJoCo, Gymnasium, SB3)

github.com/FavouritePlayer/ant_sim

- Trained PPO quadruped policies in MuJoCo/Gymnasium; terrain-adapted policy achieves **2.1x reward (893±130 vs. 424±106)** and 40% lower fall rate than the flat-trained baseline on unseen terrain, validated across 10 matched seeds per condition.
- Built leg-amputation-robust specialist policies (**~49x flat-baseline reward**) and a compound policy router that switches checkpoints by damaged leg, beating any single policy across a 4-leg × 10-seed evaluation sweep.
- Validated all headline results via **multi-seed training replication and an automated regression test suite**; generated comparison videos for each experiment.

Face Morpher — From-Scratch Warping (NumPy, dlib, scipy)

github.com/FavouritePlayer/face-morpher

- Implemented affine warping, inverse mapping, and bilinear interpolation **from scratch in NumPy** to remap pixels between corresponding faces.

Skills

Languages: Python, Java, JavaScript, SQL, Dart, Scheme

Technologies: PyTorch, Transformers (BART), scikit-learn, Pandas, NLTK, spaCy, MuJoCo, Gymnasium, Stable-Baselines3, React.js, Angular, Spring Boot, Flutter, Next.js, Firebase, AWS, MongoDB, Git.